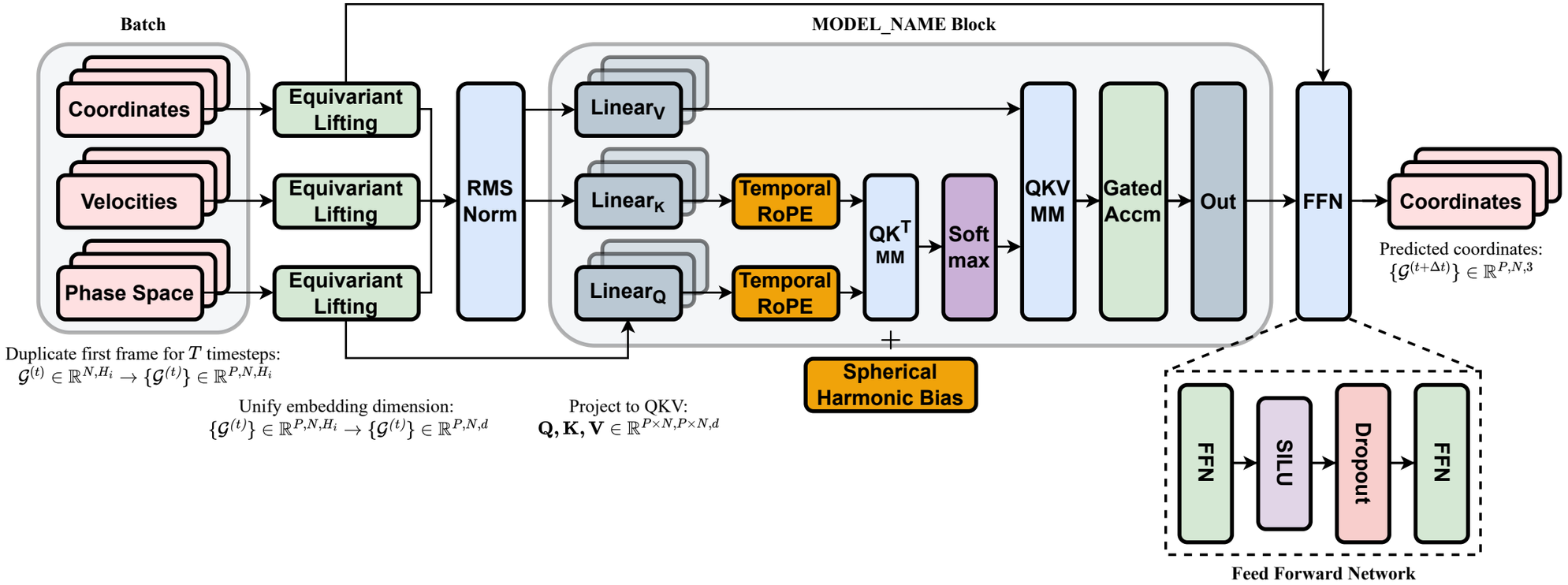
Batch
MODEL_NAME Block
Coordinates
Velocities
Phase Space
Equivariant Lifting
Equivariant Lifting
Equivariant Lifting
RMS Norm
Linear V
Linear K
Linear Q
Temporal RoPE
Temporal RoPE
QK T MM
Soft max
Spherical Harmonic Bias
QKV MM
Gated Accm
Out
FFN
Coordinates
Predicted coordinates: $\{\mathcal{G}^{(t+\Delta t)}\} \in \mathbb{R}^{P,N,3}$
Duplicate first frame for T timesteps: $\mathcal{G}^{(t)} \in \mathbb{R}^{N,H_i} \rightarrow \{\mathcal{G}^{(t)}\} \in \mathbb{R}^{P,N,H_i}$
Unify embedding dimension: $\{\mathcal{G}^{(t)}\} \in \mathbb{R}^{P,N,H_i} \rightarrow \{\mathcal{G}^{(t)}\} \in \mathbb{R}^{P,N,d}$
Project to QKV: $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{P \times N, P \times N, d}$
FFN
SILU
Dropout
FFN
Feed Forward Network